

# Know When to Fold: Risk-Controlled Abstention for Protein–Ligand Co-Folding under Novelty Shift

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## Abstract

Protein–ligand co-folding models (AlphaFold3, Boltz, Chai) report confidence scores that correlate with pose accuracy, but correlation is not a decision rule, and it is weakest on the novel pockets and chemotypes drug discovery cares about. We build `foldgate`, a training-free layer that turns frozen co-folding confidence into a risk-controlled accept/abstain gate, and audit where its guarantee fails. Calibrating on familiar targets and deploying on novel ones drives realized selective error to 0.55 (AF3, 90% CI [0.52, 0.58]; 0.48 to 0.66 across five models) against a 0.20 target. A per-stratum feasibility analysis shows this is not a tuning problem: of the 40 non-reference model-by-stratum cells, 21 admit *no* threshold on the frozen score that reaches the target at any coverage at  $\alpha = 0.20$ , and 14 of those survive an exact binomial lower bound, so abstention is the only certified action there. An exact decomposition of the realized risk explains why and yields a measurable drift diagnostic that selects the repair from a labeled probe. Group-conditional recalibration restores per-stratum control, retaining 73% of AF3 poses with the finite-sample certificate holding in four of five held-out target folds under leakage-free cross-validation, where the native gate certifies almost none, at a price we measure: to extend one stratum outward costs on the order of 80 in-stratum labels, and the novel tail admits no repair by stratification alone. A public deployment-computable proxy for the stratifier tracks the oracle on low and moderate novelty but loosens control on the novel tail, so for closed-training models the repair is deployable one stratum out and the extreme tail remains uncertifiable. The layer runs on released predictions on CPU and ships as certificate cards per model and novelty stratum.

## 1 Introduction

Co-folding models predict a protein–ligand complex and emit confidence scores (interface predicted TM-score, ipTM; ranking score; predicted aligned error). A practitioner who wants to act on a predicted pose needs a decision rule that carries an error guarantee. Conformal selective prediction supplies one: accept a pose when its score clears a threshold, abstain otherwise, and certify that the error rate among accepted poses stays below a target  $\alpha$  with high probability [7, 23]. The catch is that this guarantee assumes exchangeability, that a new target is statistically interchangeable with the calibration examples, and that breaks precisely where deployment matters, on ligands and pockets dissimilar to the training set. In drug-discovery terms the model is most overconfident on the never-before-seen molecules a program cares about, so the operational question is not average accuracy but whether a single delivered pose can be trusted.

We make three contributions, all training-free with respect to the co-folding model and all computed from released predictions on CPU. **(1)** We give a finite-sample selective-risk gate over frozen co-folding confidence, show its guarantee collapses under novelty shift, and map the collapse with a per-stratum *feasibility frontier* that decays to zero on novel strata; reliability drift attributes the collapse to concept shift (Sec. 3–4). **(2)** We prove an exact decomposition of the realized selective risk into a covariate-corrected reference and an accept-region concept gap that no training-free reweighting can move, turning it into a measurable diagnostic that selects, from a labeled probe, which repair a deployment region admits (Sec. 5). **(3)** We repair what is repairable with group-conditional calibration and a label-free worst-subpopulation certificate, match the certifier to the loss, and *price* the repair: the label cost as a design curve, a deployment-computable proxy stratifier, and the honest bottom line that the extreme novel tail is uncertifiable by stratification alone (Sec. 6). A leakage-free utility evaluation and a decision-curve analysis follow (Sec. 7).

The closest prior work is CoDrug [14], which conformalizes a *scalar* molecular property under covariate

**Table 1:** Notation.

|                  |                                                |                  |                                              |
|------------------|------------------------------------------------|------------------|----------------------------------------------|
| $L$              | loss $\mathbf{1}[\text{RMSD} > 2 \text{ \AA}]$ | $\alpha, \delta$ | risk target, failure prob. (0.20, 0.10)      |
| $s, \tau$        | score, accept threshold ( $s \geq \tau$ )      | $c$              | coverage (accepted fraction)                 |
| $v, S_v$         | novelty axis, stratum                          | $\eta_P, \eta_Q$ | source/target error conditional $P(L=1   X)$ |
| $R_{\text{ref}}$ | covariate-corrected reference risk             | $\Delta_c$       | accept-region concept gap                    |
| $D(v)$           | reliability drift at stratum $v$               | $m^*, \beta^*$   | certified min. subpop. mass, CVaR level      |
| $\rho^*$         | DRO robustness radius                          | $n_g$            | in-stratum labels                            |

shift with the same weighted-conformal tool [3] our decomposition shows concept shift defeats in this domain; we are not aware of a prior conformal treatment of co-folding pose correctness, or of training-set similarity used as the shift variable for it. Confidence-tracks-accuracy studies for co-folding virtual screening [10] rank by ipTM but supply neither a coverage guarantee nor an abstention rule, which is the gap this layer fills. A reweighting-invariant selective-risk floor of the same abstract character was shown independently and concurrently for LLM abstention [19] (arXiv preprint, June 2026); ours is its co-folding specialization, localized per novelty stratum with a matching in-stratum achievability bracket. Conformal selection [15, 16] (the latter an arXiv preprint) is complementary: it certifies which *compounds* to advance given a labeled calibration set, while this layer certifies the *pose* correctness a co-folding screen rests on, training-free.

## 2 Data and definitions

**Unit and loss.** The decision-relevant unit is a delivered top-1 pose per system with a native confidence  $s$ . The loss is  $L = \mathbf{1}[\text{symmetry-corrected ligand-RMSD} > 2 \text{ \AA}]$  against the released BiSyRMSD label, so a certified selective risk is a bound on the accepted-set error rate. We use  $L$  throughout (Table 1). RMSD is the shipped symmetry-corrected BiSyRMSD (receptor superposition, ligand-in-place); recomputed geometry is used only where noted (Table 2).

**Novelty axes.** We stratify by training-set similarity along two structural axes: a ligand axis (ECFP4 Morgan Tanimoto to the nearest training ligand) and a pocket axis (SuCOS shape  $\times$  pocket query-coverage), both released by Runs N’ Poses. Each axis is binned into quartiles  $S_0$  (familiar) to  $S_3$ , with systems that have no training analog (missing similarity, about 3% of poses, AF3  $n \approx 76$ ) forming a fifth no-analog stratum  $S_4$ . The quartile edges are fixed a priori from the similarity marginal; Table S1 lists  $n$  and the similarity range each bin spans, per model and axis. We treat structural similarity, not deposition recency, as the operative shift variable, for a reason the data forces (Sec. 3).

**Cohort.** We evaluate five models (AF3 [11], Boltz-1 [20], Boltz-1x [21], Chai-1 [12], Protenix [13]) on Runs N’ Poses [8]. Figure 1 is the reduction chain: 13,535 raw delivered poses (six models) minus the 933 Boltz-2 [22] poses used only as an ungoverned affinity comparator (Sec. 7) give 12,602 governed poses over 2,425 systems; deduplicating to one pose per (system, model) gives 11,254 independent target-labels for the five governed models. Certificates that must be per deployed model use the target-label unit, since several ligand instances of one system fail together and pooling them as independent draws is anti-conservative. The cohort is chemically diverse and not a single-family monoculture: 55% drug-like (300–500 Da) with a 24% fragment tail, 954 receptor sequence clusters (the largest covering 7% of systems), and no PDB header class above 17%.

**Deployment cost.** The native-score gate is CPU-only and consumes one released prediction per target. An optional combined score (a calibration-only gradient-boosted map from cheap per-prediction features to  $P(\text{correct})$ : confidence, interface ipTM, ensemble spread, cross-model agreement, PoseBusters validity, ligand descriptors) improves the operating point but consumes the five-sample ensemble and cross-model features, so it needs  $N$  model inferences per target. We report the native-only guarantee prominently and flag the combined score as the upgrade. The combined score’s cross-model features are moderately correlated with novelty ( $|\rho|$  up to 0.35), so it is a partly novelty-aware re-score; this is legitimate and places the *combined* score on the achievability side of the theorem below, which governs the frozen *native* score.

**Table 2:** Label provenance. The primary selective-risk results use the shipped BiSyRMSD label. The protomer trap of App. C afflicts only cross-frame *superposition*, so the cross-model danger floor is restricted to the single-chain subset; interaction-fingerprint recovery enumerates contacts *within* each structure and matches them by residue identity (seqid, resname), which is invariant to a protomer swap (homodimer copies share numbering), and its result is unchanged on the single-chain subset (gate coefficient +0.04+0.06 there vs. +0.03+0.07 on the full set).

| Result                                       | Label / geometry                   | $n$                  |
|----------------------------------------------|------------------------------------|----------------------|
| Break, frontier, repair, utility (Secs. 3–7) | shipped BiSyRMSD                   | 11,254 target-labels |
| Cross-model danger floor / consensus (SI)    | recomputed, single-chain receptors | 6,163 instances      |
| Interaction-fingerprint recovery (Sec. 7)    | per-structure contacts vs. crystal | 12,475 poses         |

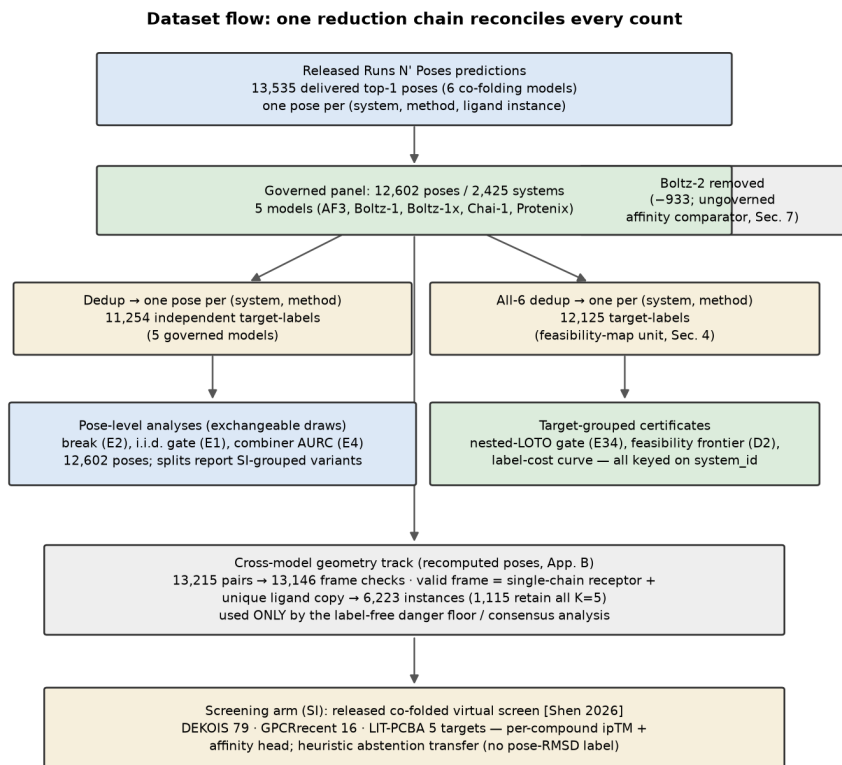
**Gate.** The gate accepts when  $s \geq \tau$  and picks the largest accept set whose selective risk is certified  $\leq \alpha$  by Learn-then-Test with fixed-sequence testing [1], testing each  $H_0(\tau) : P(L=1 \mid s \geq \tau) \geq \alpha$  with an exact binomial tail along a pre-specified, data-independent coverage grid (top- $k$  ranks,  $c \in \{0.05, 0.06, \dots, 1.00\}$ , fixed before any risk was computed and archived in the repository). This gives  $P(\text{selective risk} \leq \alpha) \geq 1 - \delta$ , finite-sample and distribution-free.

### 3 The exchangeability break

**The break.** A single global threshold hides severe per-stratum under-coverage. For AF3 the marginal realized risk  $R_{\text{marg}} = 0.177$  masks  $R_{\text{marg}, S_3} = 0.375$  on the novel ligand stratum (Fig. 2, with Clopper-Pearson intervals and accepted  $n$  per point). The operational failure is starker: calibrating the gate on the familiar strata ( $S_0, S_1$ ) and deploying on the novel tail ( $S_3, S_4$ ) drives the realized error to  $R_{\text{transfer}} = 0.55$  (AF3, 90% CI [0.52, 0.58],  $n = 734$ ; 0.48–0.66 across the five models) against the 0.20 target, so the certificate is inverted, not merely loose. We lead the break on  $S_3$ ; the no-analog  $S_4$  (AF3  $n \approx 76$ , greyed in the figure) is small-sample and enters only in the feasibility frontier, where zero coverage is the point.

**Why it breaks: concept shift, not recency.** Weighted conformal would fix this if the shift were pure covariate shift, where  $P(L \mid s)$  is stable and only the marginal  $P(s)$  moves. We test that by measuring *reliability drift*  $D(v) = \sum_{\text{bins}} w [P(L=0 \mid s, S_0) - P(L=0 \mid s, S_v)]$ , the confidence-conditional correctness gap between the familiar reference and stratum  $v$ . Drift is large on structural novelty and its 90% bootstrap interval excludes zero for every model (Protenix reaches +0.63 on pocket novelty, Chai +0.52); the +0.63 is not a sparse-bin artifact, as every confidence quintile behind it holds  $\geq 103$  novel-pocket targets and the top-confidence bin still recovers to only 0.54 correct against 0.95 on the familiar stratum. Confidence is optimistic on novel structure, the signature of concept shift. A signed accept-region decomposition makes this quantitative: for AF3 on  $S_3$  the target-risk gap of 0.328 splits into a concept term of 0.309 (90% CI [0.27, 0.35]) and a covariate term of 0.018, which Sec. 5 turns into a theorem.

**The temporal axis is a benchmark artifact, not a null result.** Earlier framings contrasted a large structural drift with a near-zero temporal drift. That contrast is real but must be stated precisely: every ‘Runs N’ Poses structure is deposited after the 2021-era models’ training cutoff, so the temporal axis ranks recency *among already-out-of-training structures* rather than in- versus out-of-training, and a flat drift there is a property of the benchmark, not evidence of temporal robustness. The one model with a genuine within-benchmark cutoff boundary is Boltz-2 (2023-06-30), which is outside the governed five-model cohort and enters only here: of its 933 poses, 77 predate the cutoff and 856 follow it, and the reliability drift across that boundary (on its *pose* ranking confidence, not the affinity head) is +0.005 (90% CI [−0.075, 0.079]), while its structural  $S_3$  break is unchanged whether keyed on the 2021- or the correctly-referenced 2023-training similarity (0.364 vs. 0.367 correct). This is a single-model observation with 77 in-cutoff targets, not a cohort result; we report it as the one available genuine temporal test and do not claim a temporal-generalization result. Structural similarity is the operative shift variable throughout.

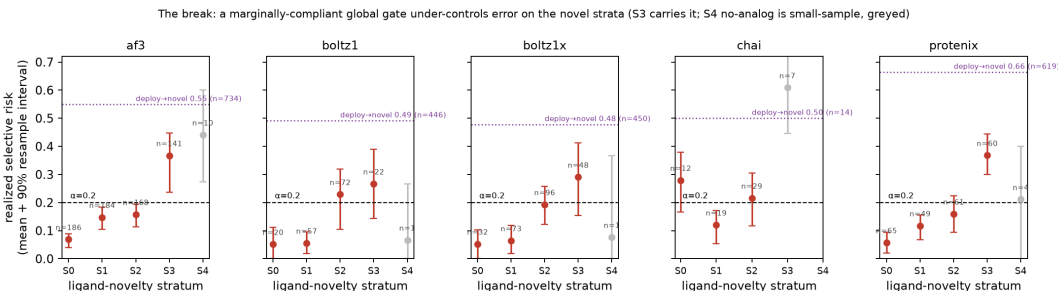


**Figure 1:** Dataset flow. One reduction chain reconciles every count; each node is annotated with the experiments that consume it. Counts are recomputed live from the released predictions.

## 4 The feasibility frontier

The break is a fixed-coverage statement; it leaves open whether  $\alpha$  is reachable by lowering coverage, so we measure that directly. We fix the deployed rule on the familiar stratum and, per target stratum, report the largest source coverage  $c_g^* = \max\{c : R_{Q,g}(\tau(c)) \leq \alpha\}$  at which it still holds  $\alpha$ , one delivered pose per (system, model). The frontier decays with novelty and then vanishes: for AF3 on the ligand axis  $c^*$  runs 1.00, 0.85, 0.75, 0.20, 0.00 across  $S_0$  to  $S_4$  at  $\alpha = 0.20$ , and 0.65, 0.40, 0.10, 0.00, 0.00 at  $\alpha = 0.10$ . Of the 50 model-by-stratum cells (two axes  $\times$  five models  $\times$  five strata), the ten  $S_0$  reference cells are feasible by construction, leaving 40 non-reference cells. Requiring at least 20 accepted targets to call a stratum feasible, **21** of those 40 admit no operating point at any coverage at  $\alpha = 0.20$ , of which **14** survive an exact binomial (Clopper-Pearson) lower bound on  $R_{Q,g} > \alpha$  at every measurable coverage; at  $\alpha = 0.10$  the counts are 26 of 40 zero-frontier and 21 robust. Pooling the two  $\alpha$  levels (47 zero-frontier cell-tests) leaves **35** robust across both levels. These are per-cell exact lower bounds; we report them as a per-test robustness check rather than a family-wise-corrected simultaneous statement, and the frontier is listed as such in the certificate ledger (Table 4). A zero frontier is strictly stronger than the coverage-pinned theorem of Sec. 5, which speaks only at the deployed coverage, and it is an empirical finding rather than a corollary. The infeasibility is not an artifact of the quartile binning: the zero-frontier fraction stays near 0.4 whether we use 3, 5, or 7 strata or fixed-similarity edges. The deployed view is starkest of all: the AF3 rule that Learn-then-Test certifies at  $\alpha = 0.20$  on familiar targets realizes risk  $R_{\text{gate},S_3} = 0.538$  on  $S_3$ , a margin of  $-0.338$ .

A one-sided escape survives the impossibility, which constrains only certified *upper* bounds: cross-model pose disagreement supplies a label-free *lower* bound on ensemble risk by triangle inequality to



**Figure 2:** The break. Per-stratum realized selective risk of a marginally-compliant global gate, over 300 target-grouped calibrate/deploy resamples (mean, 90% resample interval, median accepted  $n$  per point); the no-analog  $S_4$  is greyed as small-sample. Realized risk climbs above the target  $\alpha = 0.20$  as ligand similarity falls, and the dotted line marks the calibrate-on-familiar / deploy-on-novel realized risk (AF3 0.55). The break is carried by  $S_3$ .

the crystal pose, extending an oracle-free disagreement floor known for discrete outputs [27, 28] and the folklore that pose agreement filters wrong poses [29] to a continuous structured target. It is blind to consensus error, where all models share one wrong mode: the consensus regime covers 64% of the analyzed complexes at risk 0.152 against 0.698 on the diverse remainder, and the hidden error grows from 0.058 on  $S_0$  to 0.120 on  $S_3$  even as consensus mass shrinks. The construction and its single-chain provenance are in the SI.

## 5 An exact decomposition, and what a training-free reweighting cannot fix

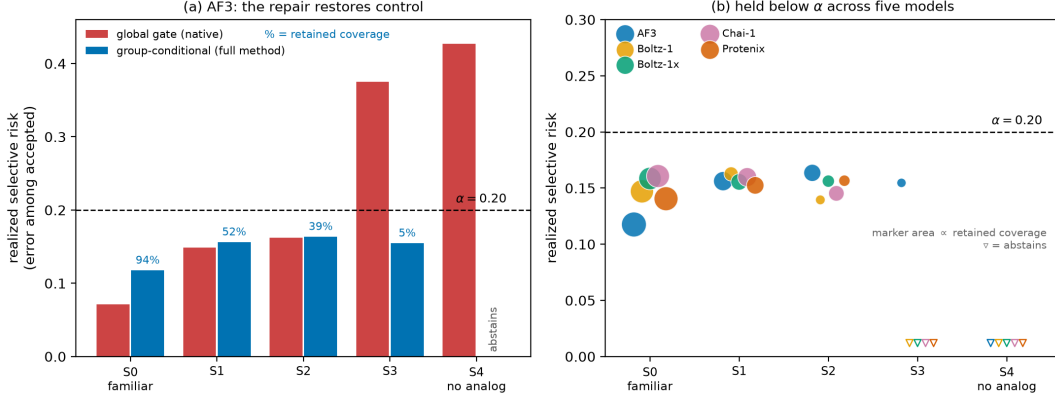
Fix the frozen score  $s$  and the accept rule  $\{s \geq \tau\}$ ; write  $Y = L$  for the error label. Let  $\eta_P(x) = P(Y=1 | X=x)$  and  $\eta_Q$  be the error conditionals on the calibration law  $P$  and the deployment law  $Q$ , and call a reweighting *training-free* if its weights are covariate-measurable and use no target labels (density-ratio, classifier, or KLIEP weights, including the point mass of weighted conformal). Write  $R_{\text{ref}}(\tau_c) = \mathbb{E}_Q[\eta_P | s \geq \tau_c]$  and  $\bar{\Delta}_c = \mathbb{E}_Q[\eta_Q - \eta_P | s \geq \tau_c]$ .

**Proposition (exact decomposition; App. A).** At fixed coverage  $c$  the realized target selective risk is  $R_Q(\tau_c) = R_{\text{ref}}(\tau_c) + \bar{\Delta}_c$ , and  $\bar{\Delta}_c$  is invariant to the choice of training-free weights. Hence the oracle-weighted plug-in certificate reports  $R_{\text{ref}}$  and under-states the realized risk by exactly  $\bar{\Delta}_c$ , and level  $\alpha$  is unachievable at coverage  $c$  by any training-free reweighting whenever  $\bar{\Delta}_c > \alpha - R_{\text{ref}}(\tau_c)$ .

The content is not that  $\alpha$  is achievable iff the risk is  $\leq \alpha$  (definitional), nor that the accept set is fixed at fixed coverage (a one-line remark), but that the accept-region concept gap  $\bar{\Delta}_c$  is *measurable and invariant*: the part of the error that comes from confidence meaning something different on novel molecules is exactly the part no unlabeled reweighting can see or remove. The proof is a tower-property identity: a covariate-measurable weight recovers only the source conditional  $\eta_P$  and cannot reach  $\eta_Q$ , completing the covariate-shift boundary of weighted conformal [3] and specializing the irreducible term of domain-adaptation lower bounds [25]; the same reweighting-invariant floor was shown concurrently for LLM abstention [19]. Matching achievability: in-stratum recalibration attains the floor  $R_{Q_g}$  given  $n_g$  stratum labels; pointwise conditional coverage being impossible [6], the guarantee is neighborhood-conditional, and its ratio functional is controlled with high probability by Learn-then-Test, whose finite-sample cost is exactly the label-cost curve of Sec. 6 rather than an asserted rate.

The decomposition’s equalities assume the score has no atom at the operating threshold. On the primary ranking\_score gate this holds exactly in the data (the ties mass at the deployed  $\tau$  is 0 for every model); on the interface-ipTM feature the boundary-randomization bracket is at most 0.7% of coverage. So the identity is exact for the deployed native gate, with a  $\leq 0.7\%$ -coverage bracket noted where ipTM is used.

**Validation.** A synthetic generator with a known  $P(Y | s, v)$  reproduces every term:  $R_{\text{ref}} = 0.273$  and  $\bar{\Delta} = 0.205$  sum to the realized  $R_Q = 0.478$ , the invariance of the realized risk across nine reweightings



**Figure 3:** The repair. Group-conditional calibration holds realized selective risk at or below  $\alpha = 0.20$  on the strata where the marginal gate over-shoots, folding to zero coverage on the no-analog tail rather than issuing a false certificate. Coverage labels are direct; the coverage cost grows with novelty.

(residual  $< 0.001$ ) is a regression test of the decomposition, and sweeping the target drift locates the coverage at which no reweighting reaches  $\alpha$ . On the public non-co-folding benchmarks elec2 (worst-block concept gap 0.36) and ACS Income (gap 0.05), a single-threshold gate under-controls the worst block, weighted conformal issues a certificate its realized risk then violates on elec2 but repairs on ACS, and group-conditional calibration certifies control by abstaining, with the elec2/ACS realized worst-block risks in App. B.

**Drift selects the repair.** Reliability drift  $D(v)$  is a label-based diagnostic that tracks  $\bar{\Delta}_c$ , so it decides, from a labeled probe rather than after a failed deployment, which repair a region admits. A small drift signals a covariate-dominated shift that weighted conformal repairs by choosing a more conservative threshold from unlabeled data (weighted conformal’s *only* lever is threshold selection; on the moderate ligand shift  $S_0, S_1 \rightarrow S_2$  it pulls AF3 realized error from 0.27 at 98% coverage to 0.19 at 70% coverage). A large drift signals a concept-dominated shift where weighted conformal silently violates its own certificate, leaving group-conditional recalibration as the only valid repair.

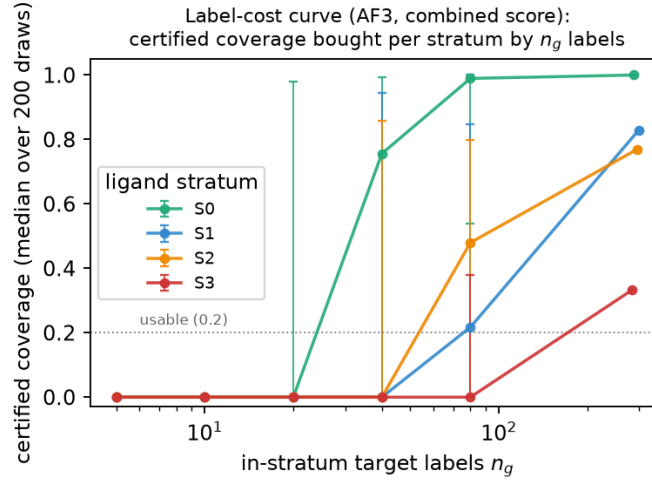
## 6 Repairing the guarantee, and pricing it

**Group-conditional calibration** [17] keys a separate threshold per novelty stratum, restoring per-stratum control at an honest coverage cost (with the combined score AF3 accepts 52% of  $S_1$  and 39% of  $S_2$  at risk  $\leq \alpha$ , against 8% and 12% natively; Fig. 3). Where the marginal gate over-shoots  $\alpha$  on the novel strata the per-stratum gate holds risk at or below it and, on the no-analog tail where no certified slice remains, folds to zero coverage rather than issue a false certificate. The repair does not require the gradient-boosted combiner: a per-stratum Venn–Abers recalibration of the native score reaches the same stratum-conditional guarantee (AF3 realized risk 0.166 at 24% coverage vs. 0.190 at 18% for the group-conditional combiner), while the field baselines break under the same shift, a fixed ipTM  $\geq 0.8$  gate at realized risk 0.30 and naive-transfer conformal at 0.41 (Table 3). A randomized localized conformal gate (RLCP) reaches lower risk at higher coverage still (0.187 at 44%), but under a weaker, randomized marginal-over-localization guarantee rather than the stratum-conditional finite-sample one we require, so we keep group-conditional as the operative repair and report RLCP as the higher-coverage option a practitioner may prefer when a randomized guarantee suffices.

**The label cost, as a design curve.** Group-conditional repair spends target labels, and how many depends on what is being certified. For the binary pose label the count is a sufficient statistic and the exact binomial tail is its exact inversion, so across the 23 feasible cells it certifies at a median of 38 independent target-labels against 60 for a Hoeffding–Bentkus bound and 102 for pure Hoeffding. But that 38 is a *certification* sample size on already-feasible (familiar) strata, not the budget to push coverage one stratum outward. Figure 4 turns the latter into a design tool: certified coverage as a function of the label

**Table 3:** Baselines under the novelty shift (AF3, calibrate on familiar, deploy on novel; realized risk / coverage at  $\alpha = 0.20$ ). The field baselines break; every gate that keys calibration on novelty holds by abstaining. RLCP reaches lower risk at higher coverage, but its guarantee is randomized and marginal-over-localization, not the stratum-conditional finite-sample guarantee this paper certifies, so group-conditional remains the operative repair and RLCP is the higher-coverage option under a weaker guarantee. Venn–Abers matches the combiner at a stratum-conditional guarantee, so the repair does not require the gradient-boosted combiner.

|                                           | realized risk | coverage |
|-------------------------------------------|---------------|----------|
| Fixed ipTM $\geq 0.8$ (field baseline)    | 0.301         | 0.69     |
| Naive-transfer conformal                  | 0.412         | 0.99     |
| Accept-all                                | 0.417         | 1.00     |
| Group-conditional (combiner)              | <b>0.190</b>  | 0.18     |
| Localized / continuous novelty (RLCP) [5] | <b>0.187</b>  | 0.44     |
| Per-stratum Venn–Abers [18]               | <b>0.166</b>  | 0.24     |



**Figure 4:** Label-cost curve. Certified coverage bought by  $n_g$  in-stratum target-labels, per ligand stratum (AF3, combined score, median over 200 draws with a 90% band). Moderate novelty ( $S_1, S_2$ ) is certifiable at a measured label budget; the near-novel  $S_3$  is label-starved even at the full pool.

budget  $n_g$ , per stratum. With the native score, no budget buys usable coverage on the novel strata, because there the rule fails, not the estimate of it, the impossibility made empirical. The combined score, a partly novelty-aware re-score and so on the achievability side, unlocks the moderate strata: AF3 reaches 22% certified coverage on  $S_1$  at  $\sim 80$  in-stratum labels and 83% with the full pool, and 77% on  $S_2$ , while  $S_3$  needs close to the full label pool for any coverage at all. The combined score raises the ceiling, not the label floor.

**Deployment-computable stratifier.** The repair needs a novelty stratum at deployment, but the oracle similarity is computed against the model’s training set, which for a closed model like AF3 a user cannot enumerate. A public proxy built from the count of pre-cutoff PDB analogs (deposition recency and protein sequence identity are at chance) recovers the oracle stratification with rank correlation 0.59 and 83% adjacent agreement, and matches oracle per-stratum control on the low and moderate strata, but loosens control on the near-novel tail (true- $S_3$  risk  $0.23 \rightarrow 0.37$ ; adjacent-stratum agreement over five bins is a lenient criterion, so this is the honest degradation). The binding limit, however, is not the proxy: the no-analog tail cannot be held even with the oracle stratifier, because its base error there is about 0.55–0.60, so a point-threshold gate keyed on the *true* stratum still realizes risk  $\approx 0.60$  and the certified Learn-then-Test group-conditional gate abstains outright (zero coverage) rather than issue a false certificate. So for closed-training models the repair is deployable one stratum out with a public

**Table 4:** Certificate ledger.  $K = 5$  models; joint claims split  $\delta$  by a Bonferroni union bound.

| Claim                     | $\delta$ | multiplicity                | $n$ (unit)     | scope            |
|---------------------------|----------|-----------------------------|----------------|------------------|
| Global / LOTO gate        | 0.10     | fixed-sequence LTT          | target-labels  | per-model        |
| Joint LOTO gate           | 0.02     | Bonferroni $\delta/K$       | target-labels  | joint (5 models) |
| CVaR / DRO bound          | 0.02     | Bonferroni $\delta/K$       | accepted poses | joint (5 models) |
| Betting bound (graded)    | 0.10     | anytime-valid (Ville)       | accepted poses | per-model        |
| Drift grid                | 0.10     | Romano-Wolf step-down       | 55 cells       | family-wise      |
| Feasibility frontier (CP) | 0.10     | per-cell exact, uncorrected | 40 cells       | per-test         |
| “For every model” effects | 0.10     | intersection-union          | per-model $p$  | no penalty       |

proxy, and the extreme novel tail is uncertifiable by stratification alone, whether the stratifier is proxy or oracle. That, not a missing training set, is the honest ceiling.

**A label-free worst-subpopulation certificate.** When the novel subpopulation is unnamed we control the conditional value-at-risk of the accepted loss, the mean error of its worst  $(1 - \beta)$ -fraction. For the binary label  $\text{CVaR}_\beta = \min(1, r/(1 - \beta))$  with accepted error rate  $r$ , so a finite-sample bound on  $r$  [2] yields a  $\text{CVaR}_\beta$  bound at every  $\beta$ , and by CVaR–DRO duality this certifies that every accepted subpopulation of mass at least  $m^* = 1 - \beta^*$  has error  $\leq \alpha$  [4], with no stratum labels and no importance weights, reparameterizing to a  $\chi^2$  robustness radius  $\rho^*$  [24]. A Bonferroni  $\delta/K$  correction gives one certificate jointly across the five models (at 20% coverage,  $m^* \leq 0.52$  and  $\rho^* \geq 0.10$ ). This certificate is a real capability on moderate shift and, we report plainly, *vacuous exactly where it is most needed*: on deploy-to-novel  $m^* \rightarrow 1$  for most models, so the label-free certificate cannot rescue the novel tail either.

**Match the certifier to the loss.** On the *graded* loss  $\min(\text{RMSD}, 4)/4$  a variance-adaptive betting bound [30] certifies a usable operating point where a distribution-free bound certifies none: AF3 certifies 43% coverage at a 1 Å mean-RMSD target with realized capped-mean RMSD 0.90 Å, against essentially zero (0.04%) for a worst-case Hoeffding bound; the empirical guarantee coverage is 0.90 (46 of the 51 non-empty gates over 120 pose-level resamples met the target; at this resample count the estimate is consistent with the nominal  $1 - \delta = 0.90$  to within its Monte-Carlo error). The reliability layer is therefore not tied to the 2 Å convention, and “1 Å” is an actionable operating point where “ $\alpha = 0.20$ ” is not.

**Certificate ledger.** Table 4 states, per claim, the  $\delta$ , the multiplicity method, the  $n$ , and whether the claim is joint or per-model, so no favorable  $\delta$  is chosen per claim. The 55-cell drift grid (5 models  $\times$  11 axis-strata: 4 ligand + 4 pocket + 3 temporal non-reference strata) uses a Romano-Wolf step-down bootstrap [31] at family-wise 0.10; all 40 structural cells survive, an effect-dominated family, and only 2 of 15 temporal cells do.

**The deliverable: certificate cards.** We package every repair verdict as a standardized card per (model, novelty stratum):  $\alpha$ ,  $\delta$ , calibration  $n$ , accepted  $n$ , realized risk, exact Clopper-Pearson risk upper bound, retained coverage, reliability drift  $D$ , the recommended repair, and a FEASIBLE / ABSTAIN verdict (Fig. 5). The pattern is the paper in one glance: familiar strata are FEASIBLE with a certified operating point, and the novel tail reads ABSTAIN, the certified action. This turns the layer from an audit into a deliverable a practitioner reads before trusting a delivered pose.

## 7 Utility, priced and leakage-free

**Ranking utility.** Every utility number is target-grouped. Under a nested leave-one-target-out protocol (outer GroupKFold on system id; within each training fold a grouped fit/calibration split, so the combiner, the threshold, and the test target are mutually disjoint), the certificate is the per-fold Learn-then-Test guarantee and its held-out validation is the number of folds whose realized risk lands  $\leq \alpha$ . On that standard the combined-score gate keeps **73%** of AF3 poses with the certificate holding in 4 of 5 folds and a pooled Hoeffding-Bentkus upper bound of  $0.192 \leq \alpha$ , whereas the native gate does *not* certify usable coverage leakage-free: at its 20% coverage only 1 of 2 non-empty folds holds and the pooled upper bound is  $0.223 > \alpha$ , and the Chai and Protenix native gates abstain outright (Table 5).

Certificate cards (ligand axis,  $\alpha=0.20$ ): FEASIBLE = certified operating point; ABSTAIN = the certified action

|                                                                                     |                                                                                     |                                                                                    |                                                                            |                                                                             |
|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|----------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| <b>af3 S0</b><br>FEASIBLE<br>cov 1.00 (n=279)<br>risk 0.13 (UB 0.17)<br>D=None      | <b>af3 S1</b><br>FEASIBLE<br>cov 0.72 (n=435)<br>risk 0.16 (UB 0.19)<br>D=0.105     | <b>af3 S2</b><br>FEASIBLE<br>cov 0.58 (n=343)<br>risk 0.16 (UB 0.20)<br>D=0.153    | <b>af3 S3</b><br>ABSTAIN<br>no certified<br>operating point<br>D=0.422     | <b>af3 S4</b><br>ABSTAIN<br>no certified<br>operating point<br>D=0.288      |
| <b>boltz1 S0</b><br>FEASIBLE<br>cov 0.73 (n=168)<br>risk 0.14 (UB 0.20)<br>D=None   | <b>boltz1 S1</b><br>FEASIBLE<br>cov 0.43 (n=234)<br>risk 0.14 (UB 0.18)<br>D=0.111  | <b>boltz1 S2</b><br>ABSTAIN<br>no certified<br>operating point<br>D=0.208          | <b>boltz1 S3</b><br>ABSTAIN<br>no certified<br>operating point<br>D=0.392  | <b>boltz1 S4</b><br>ABSTAIN<br>no certified<br>operating point<br>D=0.306   |
| <b>boltz1x S0</b><br>FEASIBLE<br>cov 0.46 (n=103)<br>risk 0.13 (UB 0.19)<br>D=None  | <b>boltz1x S1</b><br>FEASIBLE<br>cov 0.51 (n=268)<br>risk 0.14 (UB 0.18)<br>D=0.118 | <b>boltz1x S2</b><br>FEASIBLE<br>cov 0.14 (n=74)<br>risk 0.10 (UB 0.17)<br>D=0.173 | <b>boltz1x S3</b><br>ABSTAIN<br>no certified<br>operating point<br>D=0.399 | <b>boltz1x S4</b><br>ABSTAIN<br>no certified<br>operating point<br>D=0.335  |
| <b>chai S0</b><br>FEASIBLE<br>cov 0.93 (n=238)<br>risk 0.15 (UB 0.20)<br>D=None     | <b>chai S1</b><br>FEASIBLE<br>cov 0.64 (n=365)<br>risk 0.15 (UB 0.19)<br>D=0.079    | <b>chai S2</b><br>ABSTAIN<br>no certified<br>operating point<br>D=0.15             | <b>chai S3</b><br>ABSTAIN<br>no certified<br>operating point<br>D=0.407    | <b>chai S4</b><br>ABSTAIN<br>no certified<br>operating point<br>D=0.364     |
| <b>protenix S0</b><br>FEASIBLE<br>cov 1.00 (n=259)<br>risk 0.15 (UB 0.20)<br>D=None | <b>protenix S1</b><br>FEASIBLE<br>cov 0.14 (n=81)<br>risk 0.11 (UB 0.19)<br>D=0.111 | <b>protenix S2</b><br>ABSTAIN<br>no certified<br>operating point<br>D=0.245        | <b>protenix S3</b><br>ABSTAIN<br>no certified<br>operating point<br>D=0.52 | <b>protenix S4</b><br>ABSTAIN<br>no certified<br>operating point<br>D=0.437 |

**Figure 5:** Certificate cards (ligand axis,  $\alpha = 0.20$ ), one per (model, stratum). Green FEASIBLE cards carry a certified operating point (retained coverage, accepted  $n$ , realized risk with its Clopper-Pearson upper bound, reliability drift  $D$ ); red ABSTAIN cards report that no certified operating point exists, which is itself the certified action. The full grid across both novelty axes is in the SI.

So the honest leakage-free story is that the combined score is what makes the gate certifiable at usable coverage, not that it beats an equally-certified native gate; the earlier held-out pair (71% vs. 22%) is discarded because a pooled split leaks 95% of test targets into calibration where the grouped protocol shares zero. At the stringent  $\alpha = 0.10$  the combined gate certifies 13% of AF3 poses at realized risk 0.066 (pooled upper bound 0.094).

**One map for every AF3 coverage number.** Because the paper reports AF3 coverage under several protocols, Table 6 lists each with the experiment that produced it, so no two figures are mistaken for the same quantity. The 0.52-on- $S_1$  and 0.83-on- $S_1$  pair is the sharpest: the first is the coverage of a single  $\alpha$ -certified per-stratum gate, the second is the maximum coverage the full in-stratum label pool certifies (the label-cost-curve endpoint) — different questions, same stratum.

**A decision curve retires the choice of  $\alpha$ .** Rather than defend a level, we sweep the cost ratio  $\lambda = \text{cost}(\text{act on a wrong pose})/\text{cost}(\text{abstain on a right pose})$  and plot net benefit (Fig. 6). The conformal gate is best for  $\lambda \in [0.65, 4.55]$  (AF3; every model shows the same window), dominating a fixed-ipTM threshold at *every*  $\lambda$  and beating accept-all once wrong poses are more than  $\sim 0.65\times$  as costly as a missed right one. Below  $\lambda = 0.65$  accept-all wins (abstention only forfeits benefit when wrong poses are cheap), and above  $\lambda = 4.55$  abstain-all wins (no accepted set is worth its false positives), so the gate occupies the intermediate, high-but-finite-cost regime, which is where a delivered pose that must be trusted lives.

**A non-circular quality, honestly conditioned.** Accepted-pose purity is 1 – selective risk by construction, so it cannot show the gate buys anything past the guarantee. Interaction-fingerprint recovery is a different quantity a medicinal chemist reads: the fraction of the crystal’s protein-ligand contacts a pose reproduces, where a contact is any receptor residue with a heavy atom within 4.5 Å of a ligand

**Table 5:** Leakage-free nested-LOTO gate, native vs. combined score ( $\alpha = 0.20$ ,  $\delta = 0.10$ ). The certificate is the per-fold Learn-then-Test guarantee; “folds” is the number of non-empty held-out folds whose realized risk landed  $\leq \alpha$  (its validation), and  $\text{HB}^\dagger$  is the more conservative pooled Hoeffding-Bentkus upper bound. A pooled  $\text{HB}^\dagger \leq \alpha$  ( $\checkmark$ ) is a sufficient aggregate certificate; a value  $> \alpha$  means the pooled bound does not clear  $\alpha$  even where most folds hold. Chai/Protenix native gates abstain outright.

| model    | score    | cov         | risk     | $\text{HB}_{90}^\dagger$ | folds |
|----------|----------|-------------|----------|--------------------------|-------|
| AF3      | native   | 0.20        | 0.189    | 0.223                    | 1/2   |
|          | combined | <b>0.73</b> | 0.176    | 0.192 $\checkmark$       | 4/5   |
| Boltz-1  | native   | 0.21        | 0.188    | 0.222                    | 4/5   |
|          | combined | 0.60        | 0.181    | 0.201                    | 4/5   |
| Boltz-1x | native   | 0.20        | 0.164    | 0.197 $\checkmark$       | 4/5   |
|          | combined | 0.48        | 0.182    | 0.203                    | 3/4   |
| Chai-1   | native   |             | abstains |                          |       |
|          | combined | 0.64        | 0.178    | 0.196 $\checkmark$       | 4/5   |
| Protenix | native   |             | abstains |                          |       |
|          | combined | 0.57        | 0.182    | 0.201                    | 4/5   |

**Table 6:** Every headline AF3 coverage figure, mapped to its protocol ( $\alpha = 0.20$ ).

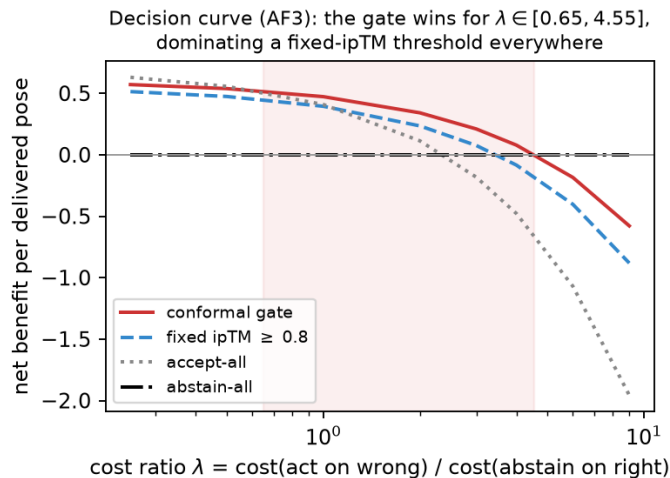
| Quantity                      | Protocol                             | Scope       | coverage    | risk          |
|-------------------------------|--------------------------------------|-------------|-------------|---------------|
| Combined gate, marginal       | nested target-grouped LOTO           | all strata  | 0.73        | 0.176         |
| Native gate, marginal         | nested target-grouped LOTO           | all strata  | 0.20        | 0.189         |
| Group-conditional (combined)  | per-stratum $\alpha$ -certified gate | $S_1 / S_2$ | 0.52 / 0.39 | $\leq \alpha$ |
| Deploy-to-novel (combined)    | calibrate familiar, deploy novel     | $S_2, S_3$  | 0.18        | 0.190         |
| Label-cost maximum (combined) | full in-stratum label pool           | $S_1 / S_2$ | 0.83 / 0.77 | $\leq \alpha$ |

heavy atom, enumerated within each structure and matched to the crystal contact set by residue identity (no interaction typing or protonation modeling). Pooling all accepted against all rejected poses shows a large recall gap (0.19 for AF3), but most of that is the accepted set being enriched for low-RMSD poses. Conditioning removes the confound: *within* the sub-2 Å set (AF3 8,277 correct poses across models) the accepted-minus-rejected recall gap is +0.02 to +0.04, and a regression of recall (a  $[0, 1]$  fraction) on RMSD with the gate as a 0/1 covariate leaves a gate coefficient of +0.03 to +0.07 recall units (all five 90% intervals excluding zero), stable at +0.04 to +0.06 on the single-chain subset. So the gate buys a small but robust non-circular interaction-recovery lift, once the RMSD confound is removed. Separately, the accepted set is PoseBusters-valid at a higher rate than the rejected set for every model (AF3 0.77 vs. 0.58), a free strengthening; certifying the joint label ( $\text{RMSD} \leq 2 \text{ Å}$  and physically valid) directly is expensive because the confidence ranks for correctness, not validity, and we report that cost rather than hide it.

## 8 Generality and limitations

**Pseudo-prospective.** Freezing calibration on everything deposited before 2023-05-31 and deploying on everything after (per model  $\sim 1300$  pre- and  $\sim 900$  post-cutoff targets), with no stratum information at calibration: on the native gate, AF3, Boltz-1x, and Protenix hold the post-cutoff certificate (realized-risk upper bound  $\leq \alpha$ ), Boltz-1 overshoots by its upper bound (0.207), and Chai abstains; on the combined gate, AF3 and Boltz-1 hold, Chai overshoots (0.202), and Boltz-1x and Protenix abstain. Abstention is a non-certification, not a violation, and no held gate’s realized risk exceeds its expected (pre-cutoff) risk by more than 0.00, so temporal deployment does not silently break control. This is a recency transfer, complementary to the structural story, and inherits the caveat that Runs N’ Poses is entirely post-cutoff.

**Cross-dataset.** Recast as risk control rather than ranking, a threshold frozen on Runs N’ Poses and deployed on regenerated FoldBench [9] Protenix predictions accepts only 3 of the 52 low-homology targets (coverage 6%, Clopper-Pearson risk upper bound 0.63), so the low-homology arm is uninformative



**Figure 6:** Decision curve (AF3). Net benefit per delivered pose vs. the cost ratio  $\lambda$ ; the shaded band is where the conformal gate is the best of the four strategies. It dominates a fixed-ipTM threshold everywhere and serves the high-cost regime, so there is no single  $\alpha$  to defend.

about risk and informative only about coverage *starvation*, the coverage collapse of Sec. 3 reproduced on a second dataset; on the 384 train-similar targets the frozen gate accepts 18 at realized risk 0.11. The interface-ipTM gate out-ranks a `ranking_score` control (AURC 0.380 vs. 0.454) but their 90% bootstrap intervals overlap, and the regenerated model’s self-scored top-1 success (0.40) differs from the released 0.57, so this stays a directional transfer check, feature and label self-consistent within the one run. A third dataset would need a benchmark that releases ipTM and PoseBusters per pose; where none is available we state that concrete community ask rather than overclaim generality.

**Limitations.** The no-analog tail is uncertifiable by every method, where abstention is the only action. The best ranker in our screening analysis (SI), a Boltz-2 affinity head, carries no coverage guarantee: extending the framework to certify an affinity ranker (which needs an affinity-error label) is the top open problem. All evaluation is retrospective; a prospective screen with wet-lab confirmation is the natural next step.

## 9 Conclusion

Co-folding confidence becomes a risk-controlled decision only when the guarantee is audited where it fails. On structural novelty a marginally valid gate that calibrates on familiar targets and deploys on novel ones realizes error 0.55 against a 0.20 target (about  $2.7\times$ ), and across both  $\alpha$  levels 35 of 47 zero-frontier cell-tests admit no threshold on the frozen score that is certifiable at any coverage. An exact decomposition explains why: the concept gap that opens there closes under in-stratum recalibration but under no training-free reweighting. Three honest bottom lines follow. Co-folding confidence is certifiable on the familiar strata (where the exact binomial certifies from a median of 38 labels over the 23 feasible cells, against 102 for Hoeffding). It is extendable one stratum outward at a measured price, on the order of 80 in-stratum target structures for usable certified coverage, and with a public proxy stratifier for closed-training models. And on the extreme novel tail, exactly where a program most needs a guarantee, neither stratification nor a label-free certificate can supply one, so abstention is the certified action. The layer runs on frozen models’ released predictions on CPU and ships as certificate cards per model and novelty stratum.

**Data and code availability.** Code and calibration tables are at <https://github.com/rihinkavuru/foldgate> under an OSI-approved license; a one-command `make repro` reproduces the tabular results from a pinned environment (uv.lock, fixed seeds), with the screening, cross-model, and structure-feature arms opt-in because they consume larger released artifacts. Runs N’ Poses is at Zenodo (10.5281/zenodo.14794785);

the regenerated FoldBench predictions and self-scored labels are at Zenodo (10.5281/zenodo.21417241); the screening scores are from (author?) [10]. The coverage grid, fixed-sequence order, binning, and endpoints are frozen in a timestamped pre-registration file in the repository.

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## A Proof of the decomposition, and the ties bracket

**Setup.** Frozen score  $s = s(X)$ , error label  $Y \in \{0, 1\}$ , accept region  $A_\tau = \{s \geq \tau\} \in \sigma(X)$ . A training-free reweighting has weights  $w = w(X) \geq 0$  that are  $\sigma(X)$ -measurable and use no target labels. Assume  $s$  has no atom at the operating threshold, so for each coverage  $c$  there is a unique  $\tau_c$  with  $Q(s \geq \tau_c) = c$ ; real scores carry ties handled by boundary randomization, under which the equalities become  $\leq / \geq$  brackets, and Sec. 5 bounds the bracket empirically (0 on the ranking-score gate,  $\leq 0.7\%$  coverage on ipTM).

**Lemma (reweighting sees only the source labels).** For any such  $w$ , the reweighted plug-in selective risk is  $\tilde{E}^w(\tau) = \mathbb{E}_P[wY \mathbf{1}_{A_\tau}] / \mathbb{E}_P[w \mathbf{1}_{A_\tau}] = \mathbb{E}_{Q_w}[\eta_P | A_\tau]$ , with  $dQ_w \propto w dP_X$ . *Proof.*  $w$  and  $\mathbf{1}_{A_\tau}$  are  $\sigma(X)$ -measurable, so tower over  $X$ :  $\mathbb{E}_P[wY \mathbf{1}_{A_\tau}] = \mathbb{E}_P[w \eta_P \mathbf{1}_{A_\tau}]$ ; normalize. The target conditional  $\eta_Q$  cannot appear.  $\square$

**Proposition.** At coverage  $c$ : (a)  $R_Q(\tau_c) = R_{\text{ref}}(\tau_c) + \bar{\Delta}_c$  with  $R_{\text{ref}}(\tau_c) = \mathbb{E}_Q[\eta_P | A_{\tau_c}]$  and  $\bar{\Delta}_c = \mathbb{E}_Q[\eta_Q - \eta_P | A_{\tau_c}]$ ; (b) any  $w$  realizing coverage  $c$  accepts exactly  $A_{\tau_c}$ , so its realized risk equals  $R_Q(\tau_c)$ , independent of  $w$ ; (c) with oracle weights the certificate of the Lemma equals  $R_{\text{ref}}(\tau_c)$ , under-reporting the realized risk by exactly  $\bar{\Delta}_c$ ; (d) level  $\alpha$  is achievable at coverage  $c$  by some training-free reweighting iff  $\bar{\Delta}_c \leq \alpha - R_{\text{ref}}(\tau_c)$ . *Proof.* (a)  $A_{\tau_c} \in \sigma(X)$ , so  $R_Q(\tau_c) = \mathbb{E}_Q[\eta_Q | A_{\tau_c}]$ ; add and subtract  $\mathbb{E}_Q[\eta_P | A_{\tau_c}]$ . (b) The no-atom assumption pins  $\tau_c$  from  $c$ ; realized risk is a functional of  $(Q, \tau_c)$  alone. (c) Lemma with  $w^* = dQ_X / dP_X$  gives  $Q_{w^*} = Q$ . (d) is (a)+(b).  $\square$  The statement is coverage-pinned and binds only scalar thresholds of a fixed score with covariate-measurable weights; re-scoring on  $v$  or a per-stratum threshold escapes it (the achievability side), and in-stratum split calibration [23] gives stratum-conditional miscoverage  $\leq 1/(n_g+1)$  regardless of drift, with the ratio functional controlled by Learn-then-Test at the finite-sample cost the label-cost curve measures.

## B Benchmark and validation details

The synthetic generator draws a stratum from a tilted  $v$ -marginal, a score logit  $z \sim \mathcal{N}(m_0 - Ev, \sigma^2)$ , and a label with  $P(Y=1) = \varepsilon + (1 - 2\varepsilon) \text{sigmoid}(\beta_0 + \beta_1 z - D_{\text{eff}} v)$ , isolating covariate shift ( $E$ ), concept drift ( $D$ ), and an error floor ( $\varepsilon$ ); every proposition quantity is available in closed form and cross-checked by Monte Carlo. The real benchmarks reduce any classifier to  $(s, y, v)$ ; realized worst-stratum selective risk at  $\alpha = 0.20$ : elec2 marginal 0.31, weighted-conformal 0.34, group-conditional 0.20; ACS Income marginal 0.23, weighted-conformal 0.20, group-conditional 0.20, the reliability drift separating the two being 0.36 (elec2, concept) against 0.05 (ACS, covariate).

## C A benchmark-integrity note for cross-model pose geometry

Recomputing pose RMSD in a common frame carries a silent trap. Runs N' Poses ships only the system's receptor chain while co-folding models predict the full biological assembly, so a chain-ordinal superposition can land the ligand in the wrong protomer of a homodimer: the backbone still fits (superposition residual  $\sim 0.3$  Å) while the ligand sits tens of Ångström away. Neither the residual nor a triangle-inequality consistency check detects it, and a distance-consistency check passed at 0/13,146 while 21% of recomputed distances and 15% of binary correctness labels were wrong. Only comparison against the shipped BiSyRMSD label exposes it; restricting to single-chain receptors with an unambiguous ligand copy raises agreement from Spearman 0.620 to 0.995 and defines the 6,163-instance subset any recomputed-geometry result in this paper uses (Table 2). We flag this for reuse; a symmetry-aware chain assignment is the general fix.